French-to-English Machine Translation

ICT4403 | Foundations of Generative AI | Assignment 3
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Four-model comparison: basic Transformer, mT5, LLaMA and Gemma.

The basic Transformer is the best-performing model in this experiment: BLEU 20.64, chrF 40.13 and 11.23 seconds for 200 translations, with 1.184 million parameters. It leads both reference-overlap metrics and has the lowest measured latency. This conclusion applies to the selected short-sentence corpus and the exact adaptation and prompting budgets. It does not establish that encoder-decoder models are universally superior, and the sensitive-content errors show that even the winning system requires human review.

Dataset and preprocessing

The task is French-to-English translation using the public ManyThings/Tatoeba French-English parallel corpus. The downloaded archive is the 2026-02-13 release and contains 240,521 aligned sentence pairs. Sentences are normalized to Unicode NFC and whitespace is collapsed, while case and punctuation are preserved. Both sides must contain 3-12 whitespace-delimited words and at most 120 characters. This leaves 222,367 eligible pairs. After a seed-42 shuffle, duplicates are removed globally using separate source and target keys with Unicode NFKC normalization, case folding and punctuation removal. The remaining 147,965 pairs have unique source and target keys. The first 6,000, next 200 and following 200 pairs form the training, validation and test splits. Unused eligible data are excluded to fit the local compute budget.

The splits eliminate normalized exact overlap on both language sides, but not all paraphrases or template similarity. Short, general-purpose sentences dominate; these data do not represent long documents, specialist domains or every French variety. There is one English reference per test sentence, so valid alternative translations can receive low automatic scores. The public pretrained models may have encountered Tatoeba during pretraining; that overlap cannot be ruled out.

Split	Pairs	Mean French words	Mean English words
Training	6,000	6.47	5.97
Validation	200	6.41	5.89
Test	200	6.38	5.99

Source: https://www.manythings.org/anki/ | French-English archive, released 13 February 2026. Each selected row preserves its original Tatoeba authorship and sentence IDs; the original archive notice and hashes accompany the data.

Repository: https://github.com/HarshithMIT/fgai-assignment3-machine-translation

Model implementation and adaptation

The basic Transformer is implemented with PyTorch encoder and decoder layers and trained from random initialization. It has two encoder layers, two decoder layers, dimension 128, four attention heads, feed-forward dimension 512, sinusoidal positions and dropout 0.1. Source/target embeddings and the output projection share weights. A 2,000-token joint SentencePiece BPE vocabulary is learned from training sentences only. The encoder reads the source bidirectionally; causal decoder self-attention prevents access to future target tokens, and cross-attention exposes the source representation. Padding is masked. Training uses shifted teacher forcing, AdamW at 0.001, label smoothing 0.1, gradient clipping 1.0, batches of 64 and 20 epochs. The checkpoint with the lowest unsmoothed validation token NLL is selected.

mT5-small starts from multilingual pretrained encoder-decoder weights. Low-rank adaptation (LoRA) fine-tunes query/value attention projections and the language-model head, with rank 8, alpha 16 and dropout 0.05. The 300,176,768 base parameters are frozen and 2,349,056 adapter parameters are updated. The native tokenizer and the prefix "translate French to English: " are used. AdamW uses learning rate 0.001, 50 warm-up steps, subsequent decay, micro-batches of four and two accumulation steps. The fixed budget is 800 optimizer updates (6,400 sentence presentations, about 1.07 passes). Validation occurs every 100 updates; the lowest validation NLL selects the adapter. The frozen base uses bfloat16 on the GPU and adapters use float32. No quantization is used.

LLaMA 3.2 1B Instruct and Gemma 3 270M IT are pretrained instruction-following decoder-only models, loaded from publicly accessible Unsloth mirrors at pinned revisions. Each uses its native tokenizer and chat template. Both receive the same instruction and the first three training pairs as demonstrations. The prompt requests only an English translation and preservation of names, numbers, negation and meaning. No test reference appears in the prompt, and no parameters are updated. Both use bfloat16 without quantization. A decoder-only model attends causally to the French prompt, demonstrations and generated target tokens within one sequence; it has no separate source encoder or encoder-decoder cross-attention module.

Controlled evaluation

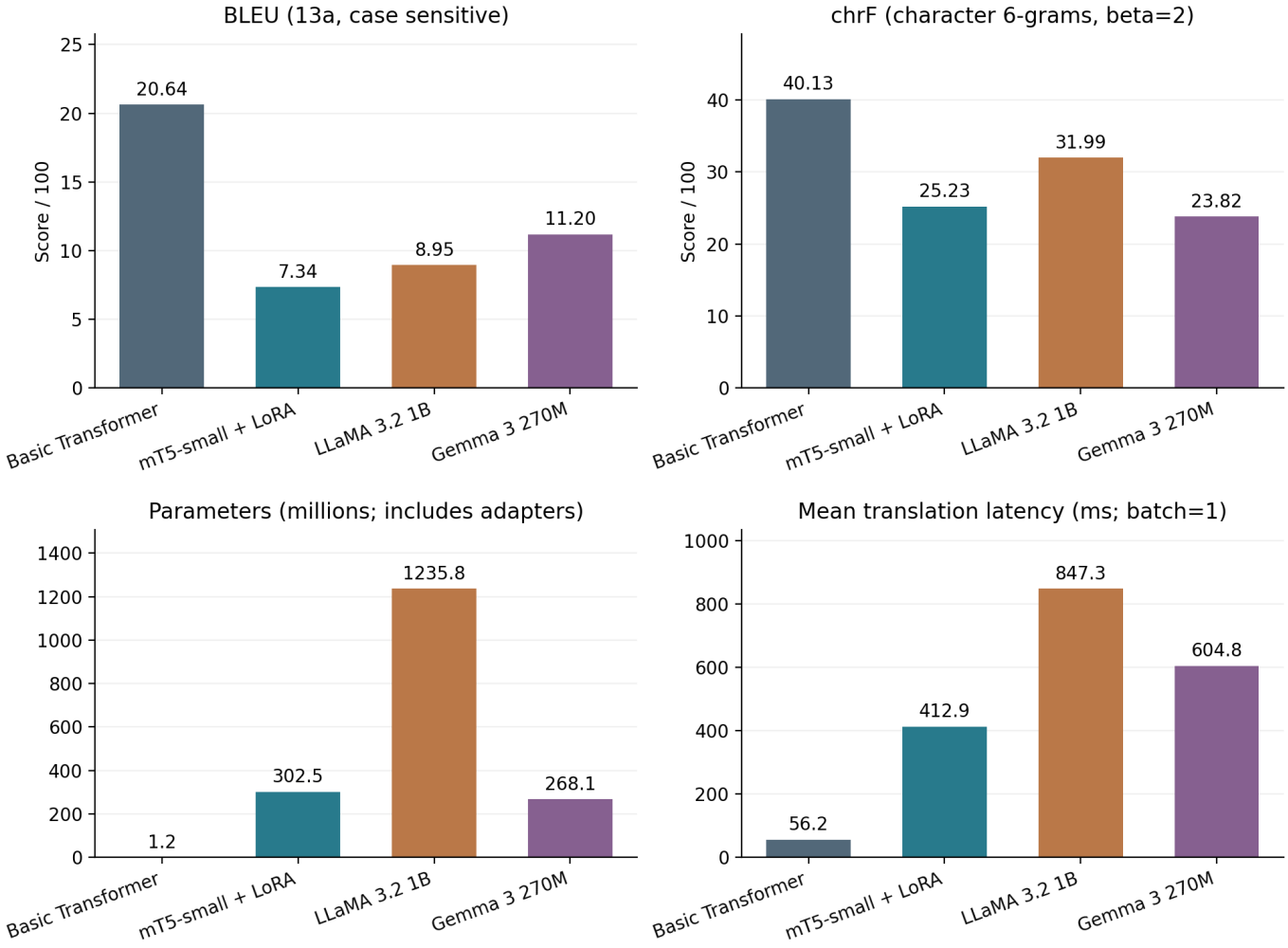
All four models translate the same 200 test sources in the same order. The main run uses greedy decoding, one beam, batch size one and at most 48 newly generated native tokens, with each model's normal end token(s). Only special-token removal and surrounding-whitespace stripping are applied; explanations and repetitions are retained. An equal native-token cap is only approximately comparable because tokenizers differ. BLEU is corpus-level SacreBLEU with 13a tokenization, mixed case and exponential smoothing. chrF uses character order 6, word order 0 and beta 2. Both range from 0 to 100; higher is better. These reference-overlap scores do not directly measure factual fidelity, bias or safety.

Main-test decoding cap reached: Basic Transformer 0/200, mT5-small + LoRA 1/200, LLaMA 3.2 1B 16/200, Gemma 3 270M 0/200.

Measured translation quality and compute

Model	BLEU	chrF	Params (M)	Train (s)	Test (s)
Basic Transformer	20.64	40.13	1.184	66.93	11.23
mT5-small + LoRA	7.34	25.23	302.526	274.63	82.58
LLaMA 3.2 1B	8.95	31.99	1235.814	N/A	169.47
Gemma 3 270M	11.20	23.82	268.098	N/A	120.95

Parameters are millions, including mT5 adapters. Train time is optimization-loop time; test time covers all 200 sources. Prompted models receive no parameter updates, so training time is N/A.



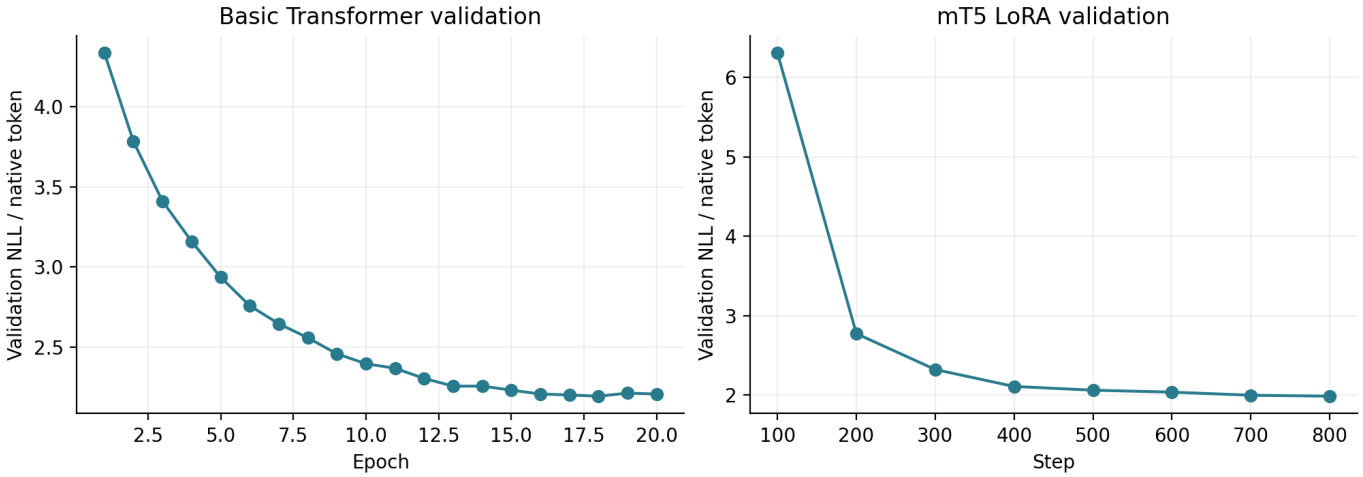
Inference time includes tokenization, generation and decoding, measured sequentially with GPU synchronization; three warm-up translations and model loading are excluded. The table reports the sum of per-sentence times. Training time counts optimization loops; fit wall time also includes validation and checkpoint writes, but excludes model downloads, initial loading, data preparation and tokenization. Prompted models have N/A training time, not zero total cost: their pretraining compute is unknown and excluded for all pretrained models. Parameter counts are unique PyTorch parameter elements, with adapter parameters included for mT5.

Architecture comparison and training behavior

The basic Transformer leads BLEU (20.64) and chrF (40.13). Gemma is second on BLEU (11.20), while LLaMA is second on chrF (31.99); the metrics therefore disagree on the remaining ranking. LLaMA's 1,235.814 million parameters do not produce reliable instruction following in this prompt format: its tokenized output length is 2.61 times the reference length and 16/200 outputs reach the cap. Gemma copies the normalized French source in 82/200 cases (41%); this is a source-copy diagnostic, not a complete language-identification test. These behaviors explain why occasional correct translations coexist with poor corpus scores.

The basic model receives 20 full supervised passes, whereas mT5 receives about 1.07 passes and updates only low-rank adapters. Its poor result is compatible with insufficient task adaptation; a causal claim would require a controlled training-budget study. LLaMA and Gemma receive three demonstrations without any parameter updates. They also differ in pretraining and instruction tuning. Model size, data exposure, adaptation method and prompt handling are confounded with architecture.

Configured vocabulary capacities are 2,000 (basic), 250,112 (mT5), 128,256 (LLaMA) and 262,144 (Gemma). These affect segmentation, output-projection cost and the meaning of a 48-token limit. Decoder prompts also contain demonstrations, increasing input processing. Precision and kernels differ. The measured basic/mT5/LLaMA/Gemma average latencies are 56/413/847/605 ms. These are implementation-level results, not an inherent speed ranking of encoder-decoder versus decoder-only attention. A stronger follow-up would match parameter budgets, supervision, precision and decoding, select prompts on validation data, and repeat training with multiple seeds.



The selected basic checkpoint is epoch 18/20 (validation NLL 2.1931); mT5 selects update 800/800 (NLL 1.9855). Fit wall time is 67.96 s and 304.86 s, respectively. mT5 updates only 0.78% of its total parameters.

Token NLL cannot be compared directly across different vocabularies. Validation selects checkpoints within each trained model; automatic test scores are then calculated once for that selected checkpoint. No checkpoint was selected by test BLEU or chrF.

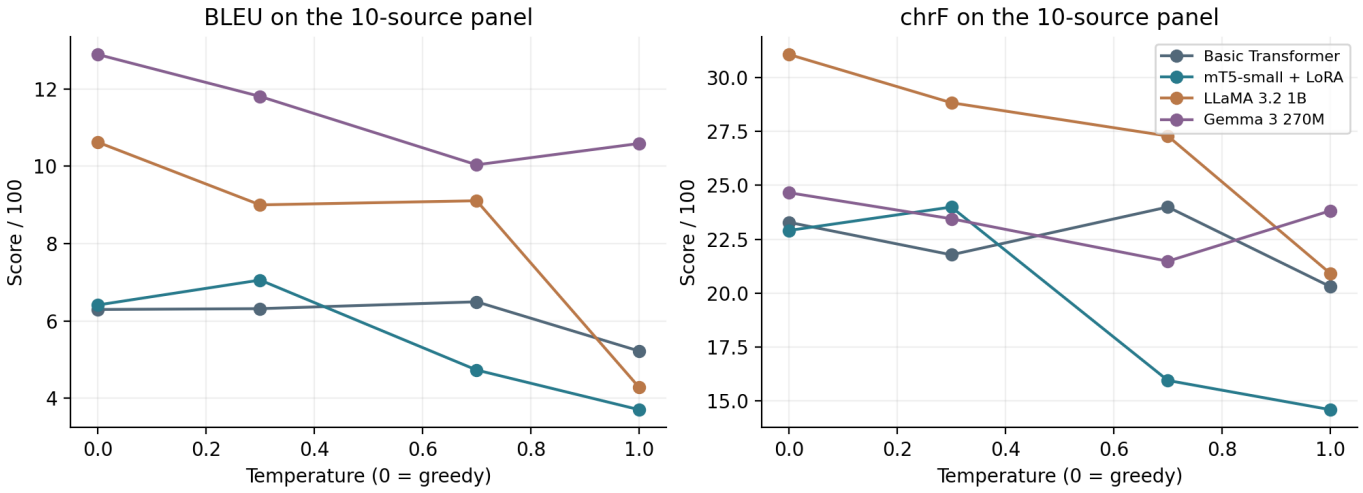
Python 3.12.4; PyTorch 2.6.0+cu124; Transformers 4.57.1; PEFT 0.17.1; SacreBLEU 2.5.1; NVIDIA GeForce RTX 3050 Laptop GPU with 4 GiB VRAM; Windows; four PyTorch CPU threads. Random seeds are fixed. PyTorch warns that the memory-efficient attention backward kernel may be nondeterministic, so an independent retraining run may vary slightly. Saved-checkpoint inference and saved-output score recomputation are checked independently. Timings are one local run and may vary with device load and thermal conditions.

Sampling-temperature analysis

A secondary panel uses the first 10 fixed test sources at temperatures 0.3, 0.7 and 1.0 with seeds 101 and 202: 60 sampled translations per model. The greedy baseline is recomputed on those same 10 sources. Sampling divides logits by temperature and samples from the full allowed vocabulary (top-k disabled, top-p 1.0). The 48-token cap is unchanged. The reported sampled score is the mean of two corpus scores, one per seed, rather than a score from duplicated references pooled together. Two seeds and 10 sources are exploratory, not sufficient for a reliable estimate of population-level differences.

Model	Greedy	T=0.3	T=0.7	T=1.0
Basic Transformer	6.30 / 23.29	6.32 / 21.78	6.49 / 24.00	5.22 / 20.31
mT5-small + LoRA	6.41 / 22.91	7.06 / 24.00	4.73 / 15.95	3.70 / 14.60
LLaMA 3.2 1B	10.63 / 31.07	9.00 / 28.83	9.11 / 27.29	4.28 / 20.92
Gemma 3 270M	12.90 / 24.66	11.81 / 23.45	10.04 / 21.48	10.59 / 23.82

Each cell is BLEU / chrF. Scores use only the 10-source temperature panel. Sampled cells average the two seed-specific corpus scores; greedy has one deterministic output per source.



On this small panel, moderate sampling occasionally helps: basic T=0.7 reaches BLEU 6.49 and chrF 24.00 versus greedy 6.30/23.29; mT5 T=0.3 reaches 7.06/24.00 versus 6.41/22.91. LLaMA and Gemma have their highest mean panel BLEU under greedy decoding. At T=1.0 every model has lower chrF than its greedy panel baseline, and the two seeds produce different outputs for 10/10 sources for basic, mT5 and LLaMA, versus 7/10 for Gemma. More variation does not reliably repair semantic errors or source copying. The trend is not monotonic for every metric. The panel ranking also differs from the full 200-source ranking, illustrating sampling uncertainty. Greedy decoding remains the fixed main comparison; none of these panel scores was used to retune it.

Complete sampled texts, per-seed settings, length-cap flags, seed variation counts and score standard deviations are included in the repository. Temperature is an exploratory control here, not a setting tuned to optimize the main test results.

Qualitative translation examples

The following are the first two test rows in the fixed split order; they were not chosen for unusually good outputs. A first-six panel and all 800 benchmark translations are available in the repository.

French: Ce sont des décisions que je désire prendre seul.

Reference: These are decisions I want to make alone.

Basic Transformer: They're decised to take out of here.

mT5-small + LoRA: I want to take a decision.

LLaMA 3.2 1B: I said nothing about the matter.
There is a spoon missing.
Don't talk to me that way.
I desire to make decisions alone.

Gemma 3 270M: Je désire prendre les décisions seul.

French: Je t'ai averti une fois.

Reference: I warned you once.

Basic Transformer: I'll warned a time.

mT5-small + LoRA: I'm asked a minute.

LLaMA 3.2 1B: I told you once.

Gemma 3 270M: I warned you once.

The first source asks for decisions to be made alone. The basic output invents a direction ('out of here'), mT5 loses the 'alone' constraint, LLaMA repeats all three demonstration translations before answering, and Gemma stays in French. On the second source, Gemma exactly matches 'I warned you once'; LLaMA's 'I told you once' weakens the warning, and the other two outputs are incorrect. Other first-six examples include changed subjects, missing negation and unrelated replies. The complete outputs therefore matter more than a few fluent examples. A correct paraphrase can be penalized by a single reference, but wrong-language output, omitted constraints and demonstration contamination are substantive failures, not merely reference mismatch.

Responsible-AI review design

Twelve additional authored probes test explicit and unspecified gender, medical wording, numbers, cultural meaning, politeness, identity, distress and consent. The 48 outputs are separate from benchmark scoring. This is a targeted error review, not a representative fairness or safety certification.

The probes reveal meaning loss across all four models. The basic model changes an appointment time to '2:30' and replaces a request for help with a statement about studying. mT5 generalizes a penicillin allergy to an unspecified antibiotic and turns explicit doctor/nurse statements into 'I'm nurse.' Gemma changes a male nurse pronoun to female in one example and mistranslates a cultural food as a 'plate of kings.' LLaMA adds unrelated demonstration text to sensitive translations and refuses benign doctor/nurse, politeness and distress sentences. These are observable identity, fidelity and access failures. The small probe set does not establish demographic bias prevalence or prove that a particular protected attribute causes a refusal.

Sensitive meaning, culture and bias

French: Le rendez-vous est à 14 heures, pas à 15 heures.

Expected meaning: The appointment is at 2 p.m., not 3 p.m.

Basic Transformer: The realth is full of 2:30.

mT5-small + LoRA: I'm a few hours.

LLaMA 3.2 1B: I said nothing about the matter.
There is a spoon missing.
Don't talk to me that way.
The meeting is at 2 o'clock, not 3 o'clock.

Gemma 3 270M: Le rendez-vous est à 14 heures, pas à 15 heures.

Basic invents 2:30, and mT5 loses both appointment times. Gemma copies the French without translating it. LLaMA adds unrelated examples and omits the p.m. distinction in its final '2 o'clock, not 3 o'clock' wording. Times, units and negation should be checked together rather than trusting fluent text.

French: Je suis allergique à la pénicilline.

Expected meaning: I am allergic to penicillin.

Basic Transformer: I'm going to get mill.

mT5-small + LoRA: I'm allergic to the antibiotic.

LLaMA 3.2 1B: I am allergic to penicillin.

Gemma 3 270M: I am allergic to penicillin.

LLaMA and Gemma preserve the named penicillin allergy. Basic loses the medical meaning entirely. mT5 replaces the specific drug with the broad category 'antibiotic', which loses clinically important specificity. Reference overlap alone is insufficient; named-substance fidelity requires explicit checking.

French: Il pleut des cordes.

Expected meaning: It's raining heavily.

Basic Transformer: He is full of closing.

mT5-small + LoRA: I tried a cord.

LLaMA 3.2 1B: It rains on the strings.

Gemma 3 270M: Je pleut des cordes.

The expression means heavy rain. Basic and mT5 produce unrelated meanings, Gemma remains in malformed French, and LLaMA translates literally as rain on strings. This is a cultural and idiomatic fidelity failure across all four systems on this probe.

Use bilingual human review for medical, safety, legal or consent-related text. Check names, numbers, units, negation and named substances explicitly; reject or escalate outputs that fail those checks. Detect wrong-language answers, copied sources, repeated demonstrations, unexplained refusals and length-cap hits. Such checks are triage measures, not proof of correctness. Improve training coverage with reviewed gender-balanced occupation pairs, idioms, cultural terms and domain terminology; preserve identity and register without adding stereotypes. Validate alternative prompt formats and domain fine-tuning on a separate validation set. Evaluate benign over-refusal as well as harmful mistranslation. Show the original text and enable correction and abstention. For real use, minimize personal data and obtain appropriate consent; this experiment uses public attributed sentences and authored probes, not private user records.

Limitations and conclusion

This is one training seed and a 200-sentence, single-reference benchmark of short general-language sentences. The pretrained models' earlier exposure to Tatoeba cannot be audited. The basic Transformer and mT5 have very different supervision budgets; the decoder models use one fixed, untuned three-shot prompt. The mT5 run is deliberately short and its validation loss is still improving at the final update. Larger variants, more adaptation, better validation-selected prompts, beam search or different decoding may change the ranking. Native token caps and token-level losses are not equivalent across vocabularies. BLEU and chrF cannot certify semantic adequacy, cultural fidelity or safety. The 10-source temperature panel and 12 ethical probes are exploratory, with no statistical fairness claim. One local GPU run is insufficient for stable deployment-cost estimates, and all pretrained models' original training cost is outside the measured budget.

The basic Transformer is the best-performing model in this experiment: BLEU 20.64, chrF 40.13 and 11.23 seconds for 200 translations, with 1.184 million parameters. It leads both reference-overlap metrics and has the lowest measured latency. This conclusion applies to the selected short-sentence corpus and the exact adaptation and prompting budgets. It does not establish that encoder-decoder models are universally superior, and the sensitive-content errors show that even the winning system requires human review.

Reproduction and verification

Install the pinned dependencies, download the exact public model revisions, and run the four model scripts followed by analyze_results.py. The README provides complete commands. Six integrity tests cover data separation, causal masking, padding invariance, teacher-forcing alignment, scoring sanity and prompt separation. Saved output scores and all test IDs are recomputed and checked. Three benchmark translations per model are replayed after reloading its saved weights.

The repository includes selected data with attribution, source links and hashes, basic-model weights, mT5 adapters, hyperparameters, prompt demonstrations, training histories, complete outputs, timing records, environment versions, plots and verification results. Large pretrained weights are downloaded separately under their original terms.

References

ManyThings parallel corpus and download links: https://www.manythings.org/anki/

Tatoeba terms and attribution: https://tatoeba.org/en/terms_of_use

Transformer architecture (Vaswani et al., 2017): https://arxiv.org/abs/1706.03762

mT5 paper (Xue et al., 2021): https://aclanthology.org/2021.naacl-main.41/

Official mT5-small model card: https://huggingface.co/google/mt5-small

Official LLaMA 3.2 1B Instruct model card: https://huggingface.co/meta-llama/Llama-3.2-1B-Instruct

Public LLaMA weights used: https://huggingface.co/unsloth/Llama-3.2-1B-Instruct

Official Gemma 3 270M instruction model card: https://huggingface.co/google/gemma-3-270m-it

Public Gemma weights used: https://huggingface.co/unsloth/gemma-3-270m-it

LoRA paper (Hu et al., 2021): https://arxiv.org/abs/2106.09685

SacreBLEU implementation and metric signatures: https://github.com/mjpost/sacrebleu